



Data-driven modeling for the dynamic behavior of nonlinear vibratory systems

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Abstract Accurate modeling of the mapping relationship between the external excitation and the dynamic behavior of nonlinear vibratory systems is the basis for structure design, control, and optimization of vibratory systems. However, modeling the dynamic behavior of nonlinear vibratory systems with either approximate theoretical methods or numerical simulation is difficult and time-consuming due to the randomness of external excitations forced in the nonlinear vibratory systems. In the paper, an accurate and efficient model for predicting the dynamic behavior of the nonlinear vibratory system is proposed based on data-driven technology. Firstly, the datasets, consisting of the training data and validation data of the data-driven model, are obtained by traditional quantitative analysis methods, simulation approaches, or vibration tests. Then, the dependency features between the training data are extracted through a gated

recurrent unit (GRU). The mapping relationship between the dependency features and the dynamic behavior of the nonlinear vibratory system is constructed through the fully connected layer. Finally, the accuracy of the established data-driven model is assessed by three evaluation metrics (the maximum error, root-mean-square error, and goodness-of-fit index) of the machine learning. The effectiveness of the proposed data-driven model is verified through two examples, a single-degree-of-freedom Duffing equation, and a double-layer X-type vibration isolation system. The results indicate that the GRU data-driven model, which is highly consistent with the theoretical and numerical values, has high accuracy, effectiveness, and stability in identifying the dynamic behavior of nonlinear vibratory systems. The established data-driven model probably has potential applications for modeling impact isolation, vibration damage detection, and microscopic techniques.

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1 Introduction

Vibration is a common physical phenomenon in nature and engineering, and vibratory systems identification